

JAI KUMAR DEWANI

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SUMMARY

Senior .NET engineer with 5 years in high-reliability, safety-critical software at Carl Zeiss. Strong foundations in systems design, concurrent programming, and CI/CD. Bringing the rigour of medical-device development to product engineering teams.

EXPERIENCE

Carl Zeiss - Vision Care R&D and Technologies

Bengaluru, Karnataka, India

Senior Software Design Engineer

Jan 2026 – Present

- Leading Refraction Platform architecture and feature roadmap, ensuring extensible and scalable long-term platform design.
- Built an async event-queue command-coalescing system to merge high-frequency user interactions, cutting hardware execution time by **90%** and improving UX.
- On-boarded developers to the Refraction Platform codebase, designing ramp-up guides and pairing sessions that reduced time-to-first commit from 6 weeks to 2.

Software Design Engineer

July 2022 – Dec 2025

- Owned VISUCORE R1.1 delivery end-to-end, coordinating across **4 cross-functional teams** (software, testing, product, hardware).
- Implemented hybrid encryption for token exchange with the Password Manager, enabling secure credential handling across Zeiss Vision devices in a **zero-trust environment**.
- Designed versioned ASP.NET Core REST APIs with JWT authentication, backed by EF Core and MySQL, serving the Angular frontend for Password Manager System.
- Designed the asynchronous Refraction module with mutual-exclusion and atomic command guarantees, improving workflow performance by **28%** and reducing concurrency-related bugs.
- Built a C++ interop wrapper that stabilized hardware communication and reduced integration effort by 35%.
- Applied SOLID principles and DI patterns during .NET 8 migration, decoupling **30+ modules** and improving startup time by **22%** through lazy dependency resolution.
- Built reusable Angular component library with RxJS-driven data streams and observable-based state management for real-time hardware measurement display.

Software Engineer

June 2021 – June 2022

- Refactored legacy hardware command layer using the Command and Chain-of-Responsibility patterns, introducing atomic execution guarantees and eliminating race conditions across concurrent workflows.
- Improved engineering velocity by upgrading mocking infrastructure, cutting test runtime by 80% and reducing reliance on 3rd-party services.
- Integrated gRPC for IPC between services, reducing coupling and enabling parallel development across modules.
- Maintained and extended Azure DevOps CI/CD pipelines with MSBuild, SonarQube static analysis, and Blackduck dependency scanning gates, enforcing quality standards on every merge.

EDUCATION

International Institute of Information Technology, Naya Raipur

August 2017 – June 2021

Bachelor of Technology in Computer Science

Naya Raipur, Chhattisgarh India

TECHNICAL SKILLS

Languages: C#, Python, JavaScript, TypeScript

Frameworks: .NET 8, .NET Framework, ASP.NET, WPF, NUnit, gRPC, Angular

Developer Tools: Azure, MSBuild, MySQL, SonarQube, Blackduck, IRIUS Risk, Docker, GitHub Actions, CI/CD

ACHIEVEMENTS

- Recognized with **Ownership Spirit** award 2025 for leading resolution of high-impact customer blocking issues.
- Speaker at FOSS Conference on Zeiss's open-source strategy; presented gRPC adoption patterns to Zeiss-wide developer audience via internal Community of Practice.
- Contributed to open-source projects including *Excalidraw* and *Microsoft PowerToys*.
- Mentored and judged 100+ teams at national hackathons including HackCBS, HackOn, and DotSlash.
- Qualified **ACM ICPC Regionals** (Top 100 Nationwide across 2019–2020).